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The Executive Editors
Results in Engineering
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Dear Editors,

We submit the manuscript “**Uncollapsed and Metrically-Scaled 3D Reconstruction from Monocular Endoscopic Video via Pseudo-3D Heightfields**” to *Results in Engineering* as a Research Article.

Engineering problem and fit. The work addresses a concrete engineering system problem: how to build a computational pipeline that converts monocular endoscopic video into a three-dimensional surface that is simultaneously *renderable* (un-collapsed, well-formed geometry) and *measurable* (metrically scaled in millimetres). State-of-the-art feed-forward reconstruction models (VGGT, DUST3R, MAST3R, MUST3R) collapse into flat slabs or cones and lose metric scale on endoscopic footage, and classical structure-from-motion drifts or mis-decomposes under small planar baselines. Our contribution is an end-to-end geometric signal-processing system whose central object—a bilateral gradient-anchored pseudo-3D heightfield—is engineered to serve two roles at once: a *geometric matching substrate* (frame registration, dense epipolar matching, multi-stride bundle adjustment on SE(3)) and an *operator corrector* (a physical normal field that vetoes the spurious homography branch, and a calibration gauge that injects true metric scale). The system runs on CPU in its classical mode and is validated against nine baselines on four datasets.

Headline engineering results.

- **Un-collapsed geometry:** the only *monocular* reconstruction that stays Normal 3D on the benchmark ($\lambda_3/\lambda_1 = 0.42$; all feed-forward and SfM baselines ≤ 0.22).
- **Near-ideal metric scale:** $1.11\times$ recovered scale, versus $3.7\times$ – $647\times$ drift for the baselines.
- **Leading depth accuracy:** the calibrated stereo variant attains the best AbsRel (0.0285) on the public EndoNeRF benchmark, ahead of FoundationStereo (0.0321).
- **Drift suppression:** on a 17-sequence benchmark the pattern-free multi-view BA outperforms ORB-SfM (35.52 vs. 43.72 mm mean ATE) and beats VGGT-1B in a smoke-filled cavity (24.86 vs. 37.25 mm); with the gauge visible it reaches the 4.49 mm pattern-localisation floor.

Declarations. This manuscript is original, is not under consideration elsewhere, and all

authors have approved the submission. The study uses public datasets and in-house bench-top phantom recordings and involves no new human-subject or animal experiments. The authors declare no competing interests. CRediT contributions, funding, data availability, and a generative-AI declaration are included in the manuscript. Code, evaluation scripts, and result files are openly available at <https://github.com/liranyang123456-commits/re3d-heightfield>.

We believe the manuscript’s emphasis on a complete, validated engineering system for robust 3D reconstruction fits the scope of *Results in Engineering*, and we thank you for considering it.

Sincerely,

Ranyang Li, et al.